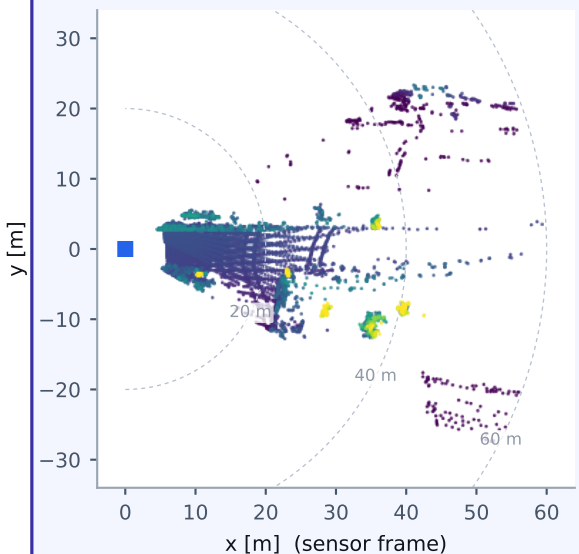


One interface, four front ends

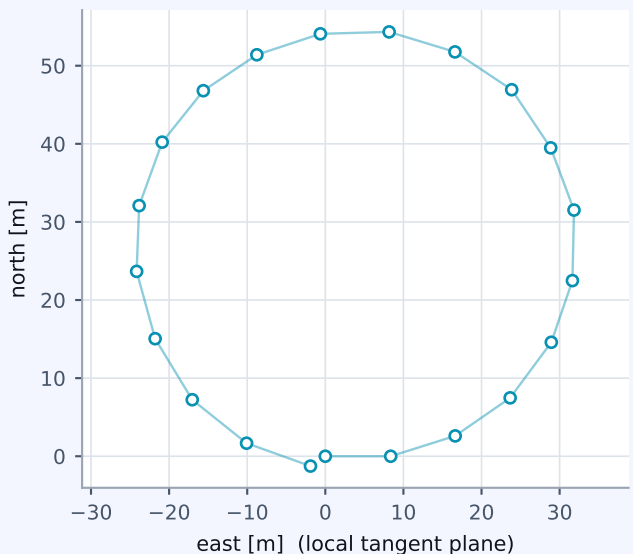
Each tinted box is one drive. Inside a box: what the sensor natively measures (left two), the trajectory that follows from its twist (third), and the twist itself (right) — the only thing the calibrator reads.

A2D2 · drive 20180810_150607

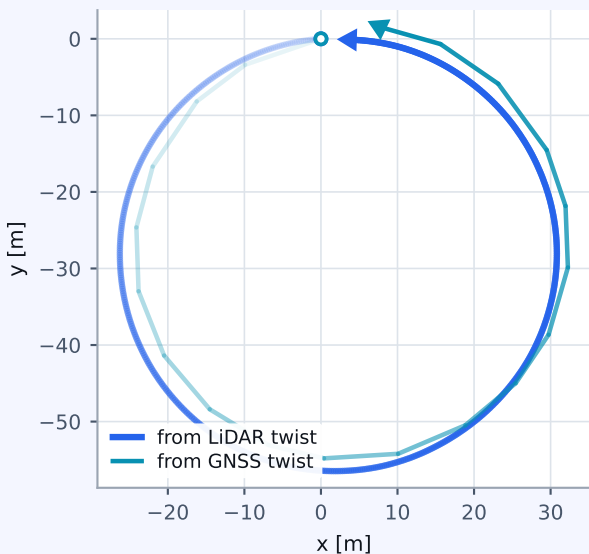
LiDAR point cloud paper front end
one scan, 10428 points, forward wedge



GNSS fixes same interface
21 distinct fixes at 1.0 Hz, no attitude



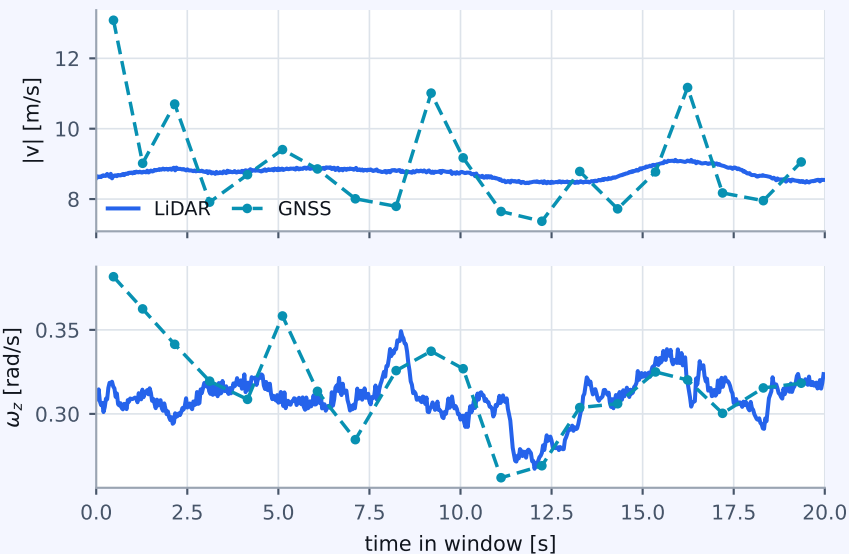
Trajectory from (v, ω)
175 m driven, 354° of heading



front ends: LiDAR (KISS-ICP, FRONT_CENTER) + GNSS fixes (vehicle bus)

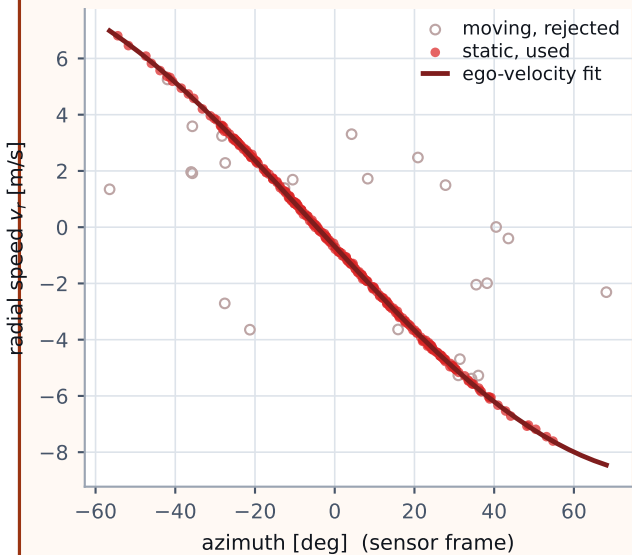
The twist, as delivered

path speed 8.75 vs 8.82 m/s · median ω gap 0.0148 rad/s

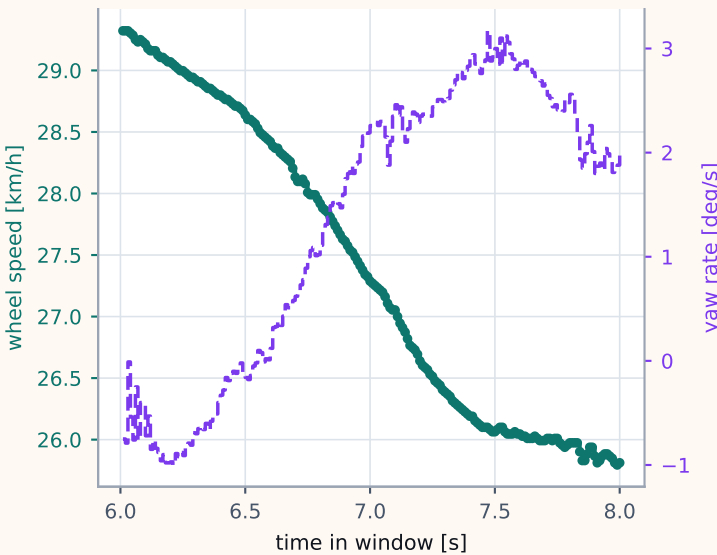


RadarScenes · sequence 1

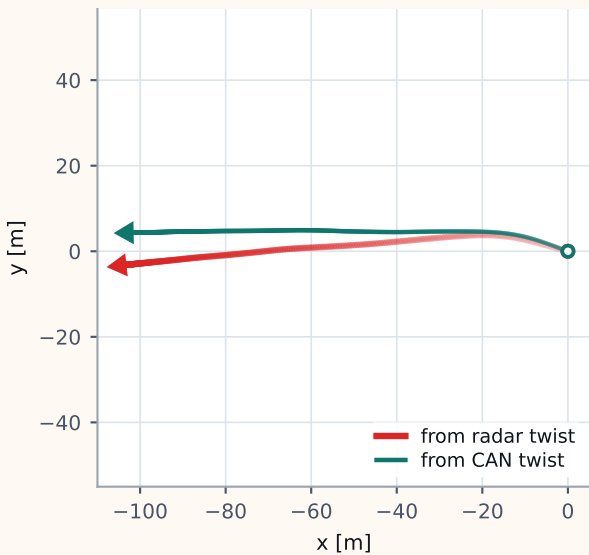
Radar detections paper front end
one scan, 310 returns, 92% inliers, residual 0.036 m/s



CAN bus frames same interface
2 s zoom of the 2000 messages in the window



Trajectory from (v, ω)
105 m driven, 22° of heading



front ends: 2-D radar Doppler (sensor 1) + CAN odometry (vehicle bus)

The twist, as delivered

median |v| gap 0.078 m/s · one 2-D radar gives no ω

